

Manual

Process Data Management PDV-PDM 8.3

Version 1.0.23049

Last changed on: 02.09.2020

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1 Overview: Process Data Management

Overview

Purpose

This product includes the graphic user interface to define master data for the Process Data Collection. The service to store the collected measured values is also included. You can implement a collection of TAG-based machine collection processes in the Process Data Collection.

To control the processing of measured values in the applications, you require the product PDV-VRP "Processing rules for process data".

Integration

This function package requires a machine interface for data collection in the Process Communication Controller (PCC).

Features

This product provides a great number of configuration functions to create and edit master data for process values.

- Configuration of logical channels to collect process values
- Definition of input and output channels
- Extensive functions to create and edit machine and article-related monitoring parameters for process values (lower/upper action limits, lower/upper tolerance limits,...)
- List to manage specific monitoring parameters for specific machines, articles and tools
- Collection of TAGs to identify measured values. Optimized storage of related measured values. The machine is used to specify interrelationships between TAGs irrespective of the time of collection. The interrelationships are saved with the identification TAGs.
- Service to index the measured values according to the defined identification TAGs.

2 Logical channels

Overview

Menu	Master data → Process data processing → Logical channels
Transaction code	lgchcnf
Function authorization	lgchcnf.*

The logical channels form the central point where logical configurations, physical machines and shop floor servers (terminal type PCC) come together.

Purpose

You must configure the logical channels in order to assign the process parameters/PDV events:

- to machines
- to shop floor servers (PCC)
- to physical channel numbers.

The logical channels define which technical channel is used on a terminal to record the characteristics.

Logical channels specify the "channel mapping".

You can also use this configuration to specify the physical channels in more detail, for example:

- you can specify how data is collected
- you can define outputs e.g. for alerts or
- you can define target values.

As of PDV 8.2, you can also configure the channels for the recording of TAGs (e.g. serial number, etc.).

This way, you can connect the measured values to an alphanumeric TAG and you can perform analyses based on TAGs in defined evaluations.

Logical channels are versioned master data. For each data record, a validity period is therefore specified (exact to the second). Only in the period specified, the data record is valid for the system.

Note: A modification of the channel configuration is not passed on to the terminal, i.e. the new channel configuration is made available to the terminal only after the terminal/ shop floor components are restarted or after the cyclical start of the configuration monitor service (e.g. in certain situations such as target value modifications).

Integration

The logical channels are used to assign process parameters or PDV events to a machine-shop floor server combination.

Selection criteria

The application provides the following selection criteria:

"General" tab

Channel

Selection of channel number

Designation

Selection of channel designation including wildcard function and search screen.

Channel type

Selection of channel type using selection box

Data class of the channel

Selection of data class using selection box

"Assignment" tab

Machine

Selection of machines including wildcard function and search screen

PCC terminal

A selection of terminal/shop floor server is possible. A search screen may also be used.

"Activation/alert" tab

Active

Selection of active channels only

Alert

Selection of logical channels with enabled option *Alert*

Valid from-to

Start and end time including date and time to specify the validity period

Alert channel

Selection of the logical channels where the respective alert channel is stored.

Field descriptions

Tab "General"

Channel

Channel number on the machine. This number must be unique for the PCC terminals. Only values between 1 and 9999 are permitted.

Designation

Optional, logical designation of the logical channel

Channel type

Includes the reference to the data that is collected via the channel. This can either be a configured PDV event (E) or a process parameter (PP, corresponds to the decimal value or the alphanumeric tag) of a characteristic.

Data class of the channel

For channels of type PP, the data class of the channel defines which value of a process parameter is collected using the channel. The values supported are:

- MV: Measured value/ alphanumeric tag
- TV: Target value
- UTL: Upper tolerance limit
- UPAL: Upper process action limit
- LPAL: Lower process action limit
- LTL: Lower tolerance limit



If a target value or a tolerance or process action limit is passed via a channel, it must be guaranteed that a time span of at least 30 minutes is respected between two value changes of a channel.

Data type (available as of PDV 8.2)

Data type that is recorded via the channel. The following three options are available:

- Decimal
- Alphanumeric
- Tag (alphanumeric)

The data types *decimal* and *alphanumeric* are used for the typical process parameters. The tag values recorded are available as selection criteria in the selection criteria based on tags. These values can be selected.

The permitted characters for the data types are defined as follows:

- Decimal: Numeric decimal values
- Tag (alphanumeric): 0-9 a-z A-Z äöüÄÖÜ _-+#,.,

Note:

The recording of tags is limited to 50 characters. If the data source transfers a tag with more than 50 characters (alphanumeric), the shop floor client cuts off the values after 50 characters.

Tab "Assignment"**Machine**

Assignment of the machine

PCC terminal

Assignment of the PCC terminal

Process parameter

Assignment of a process parameter if the channel type is defined for the collection of a process parameter.

Event

Assignment of a defined event if the channel type is defined for the collection of an event.

Tab "Properties"**Input type**

The input type of the channel is used to define how the shop floor server is to access the machine data. The control parameters are used for the collection of process values and of events (see parameter type). The values supported are:

- A: Automatic (if supported by driver)
- T: Trigger controlled
- C: Cyclic

Cycle time

If the channel is controlled using a cycle time, this field contains the number of seconds after which the values are queried.

Trigger

If the channel is controlled via trigger channel, the channel number of the trigger is stored in this field. Only values between 1 and 9999 are permitted.

Direction

Direction of data flow for the channel. Values can be transmitted from the machine to the data collection (input, column value $\uparrow$). This is the direction normally used to record machine data. In addition, the data direction can be configured from the data collection to the machine (output, column value $\circ$) to overwrite a value depending on the selected data class of the channel. For example, setting a new target value for the connected machine.

Tab "Activation/alert"**Active**

The logical channel is switched to the active status here. In data collection, only active and valid channels are used. The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Valid from-to

Validity period during which the channel is used for data collection.

Alert

This option specifies if a configured alert channel is activated (output signal for setting a physical signal). The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Alert channel

This option specifies the physical channel number where an output signal is set when an alert condition is fulfilled. Depending on the channel type, the alert can be triggered by a PDV event or a limit violation of a process parameter. Only values between 1 and 999 are permitted.

Configuration of an alert channel

In order to activate an output signal in case of a limit value violation of a process parameter, an output channel of the type "Process parameter" must be created.

The following settings are made in the dialog "Insert logical channel".

- For the channel number, you configure a "virtual channel" with any value (between 1 and 9999). However, this number must be unique. This channel number is used to manage the combination "Alert channel – data class of the channel".
- You use the "Data class of the channel" to define the limit value violation that triggers the alert.
- You must also specify the process parameter that is monitored.
- Activate "Output" as direction.
- Enable the options "Channel active" and "Alert active".
- The alert channel must match the physical channel where an output signal is set in case of an alert condition.

For a detailed description of individual input fields, refer to section "Field description".

Checking Business Parameter Containers (BSCs)

See [here](#) for further information on checking the system with respect to business parameters.

3 PDV Characteristics Master Data

Overview

Menu	Master data → Process data processing → PDV characteristics master data
Transaction code	chrp
Function authorization	chrq.*

You use the catalog of characteristics to predefine characteristics that are then used in collection rules. It aims at persons involved in data collection planning. The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options.

The screenshot shows the 'PDV characteristics master data' application window. The window has a title bar 'PDV characteristics master data' and a menu bar with 'Main page' and 'Profile'. Below the menu bar is a toolbar with icons for 'Request data', 'Cancel', 'Print preview', 'Print all', 'Insert', 'Copy', 'Edit', 'Delete', 'Save', and 'Help on operation'. The main area is divided into two panes. The left pane, titled 'Selection', has tabs for 'Main page' and 'User fields'. It contains input fields for 'Characteristic no.', 'Characteristic designation', and 'Process parameter'. The right pane, titled 'PDV characteristics master data', has tabs for 'Characteristics', 'Specifications', 'Storage', 'Visualization', 'Inspection - computation', 'Administration', and 'User fields'. The 'Characteristics' tab is active, showing a table of characteristics and a form for editing a specific characteristic.

Characteristics	Character...	Process pa...	Formula	Interval type	Interval va...	Unit
AirHumidity	Air Humidity	AirHumidity		None		
BeltVelocity	Belt Velocity	BeltVelocity		None		
CycleT	Cycle Time	CycleT		None		
CylinderTemp	Cylinder Te...	CylinderTemp		None		
DosingT	Dosing Time	DosingT		None		
Inject.T	Injection Time	Inject.T		None		
InnerPress	Inner Press...	InnerPress		None		
InnerTemp	Inner Temp...	InnerTemp		None		
MoldTemp1	Mold Tempe...	MoldTemp1		None		
MoldTemp2	Mold Tempe...	MoldTemp2		None		
NozzleTemp	Nozzle Tem...	NozzleTemp		None		
OuterTemp	Outer Temp...	OuterTemp		None		
PartsWeight	Parts Weight	PartsWeight		None		

The right pane shows the 'Characteristic' form with the following fields:

- Characteristic no.: AirHumidity
- Characteristic designation: Air Humidity
- Process parameter: AirHumidity
- Formula:

Purpose

The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options. Only enter data in the characteristics catalog that need not be changed when later assigned to a collection rule. Do not define limit values, for example. This is normally not useful because the limit values are only known when the collection rule is created. Only when you create and assign a collection rule, a relation to a concrete article or machine is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the collection rule later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a small number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article or machine only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that you can still change any definition stored in the catalog or add missing definitions when you later plan the collection rule.

Important

The configurations made in the characteristics' master data need not be definitive for the collection rule. The characteristics' master data is used as a template when the collection rules are later created. You can complete and change all configurations of the characteristics' master data in the collection rule.

Integration

The characteristics' master data is used as a template when the collection rules are later created. The field *Process parameter* provides a logical connection to the logical channels. The connection is only established when the characteristics are definitively defined in the collection rules.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Characteristic number
- Designation:
Name of the characteristic
- Process parameter:
Process parameter

If several selection criteria are used, the overlapping results are displayed below in the *Characteristics master data*.

Field descriptions

Find below a description of the columns and input options for characteristics:

Tab *Characteristics*

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the characteristic

Process parameter

Predefinition of the process parameter

Formula

See section "Formula calculation" below.

Tab *Specifications*

Select the *Specifications* tab to enter the sampling scheme and the tolerance limits. Note: As mentioned above, it is only reasonable to define tolerance limits in the characteristics master data under certain circumstances.

Sampling scheme (no longer relevant as of PDV 8.3)

The following sampling schemes are available:

- None
- Piece-related

With a piece-related sampling scheme, you can define the interval. All single values collected in this interval are then combined and form one sample. If you define a piece-related sampling scheme, you can make evaluations for samples, for example control charts.

Interval value (no longer relevant as of PDV 8.3)

Interval used to combine the single values collected in this interval to a sample.

Display of failures in the Failure Mode Analysis (PDV 8.1 and PDV 8.2)



To display the created failures in the Failure Mode Analysis, you must specify an interval value. Process parameters without specified interval value are not integrated in the Failure Mode Analysis.

**Field visibility**

The field is only visible and input is only possible if the user is assigned the function authorization "InspectionInterval".

Unit

Pieces, meter, kg, etc. Units are assigned using the unit catalog.

Decimal places (no longer relevant as of PDV 8.1)

Enter the decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Validation and tolerance limits can be entered as absolute, relative or percentage values. Note: You must enter relative or percentage limits as negative values.

Upper PL

Specifies the upper process limit. In PDV, this value also defines the displayed upper red area in process visualization.

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower process limit. In PDV, this value also defines the displayed lower red area in process visualization.

Upper TL – Generate failure / Lower TL

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:TG> or AUTO:TG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Specifications process**Upper PAL**

Upper process action limit

Lower PAL

Lower process action limit

Generate failure

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:PEG> or AUTO:PEG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Storage (no longer relevant as of PDV 8.3)

Using the filter functions for storage, you define the frequency used to store single values. This value affects the further processing of measured values on the HYDRA server. In later evaluations, only the actually stored values can be displayed. The storage frequency is also the basis for further aggregations by the PDV Distributor calculating samples. Here, the PDV Distributor only accesses data of the online data set.

Example: The defined filter specifies that only one of ten measured values is saved. At the same time, a sample interval specifies that 5 measured values are combined to one sample. After filtering, the storage frequency is the basis for the calculation of samples. 5 measured values are then actually combined to one sample. Result: For 50 measured values, which have actually been collected, one sample is generated.

Filter function:

Define a filter function for the collection of PDV values. The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*

No filter, the system saves each collected value.

- *Frequency*

Enter a number of measured values. This means that the system only saves every n measured value. The system evaluates the measured values collected in the meantime but they are not saved.

- *Cyclic*

Enter a time interval in seconds. This means that the system only saves a measured value every n seconds. The system evaluates the measured values collected in the meantime but they are not saved.

- *Percentage*

Enter a time interval and a percentage. In the PDV data collection, the system saves the

measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is saved because of the percentage deviation. The system evaluates the measured values collected in the meantime but they are not saved.

Visualization

Visualization

Check this option to enable the online visualization for the process parameter.

Position

Use this integer field to specify the visualization position of the process parameter. Use this for graphics in the online visualization with more than one display element on one page. Example: If 16 display elements are shown on one page, you can use this option to define the position of the different process parameters.

If the value 0 or less is specified, the process parameter is not visualized. You can only edit the input field after you have checked the option *Visualize*.

Visualization *Filter function*

Here, the same filter functions are available as for the storage, but you can use this second setting to decouple the storage from the online visualization. For example, you can configure that each measured value is stored, but only every tenth measured value is visualized.

The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*
No filter, the system visualizes each collected value.
- *Frequency*
Enter a number of measured values. This means that the system only visualizes every n measured value.
- *Cyclic*
Enter a time interval in seconds. This means that the system only visualizes a measured value every n seconds.
- *Percentage*
Enter a time interval and a percentage. In the PDV data collection, the system visualizes a measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is visualized because of the percentage deviation.

Inspection – computation

Check characteristic (no longer relevant as of PDV 8.3)

You can use this option to define whether the process parameter is checked against limit value violations in the PDV data collection or whether the collected measured values only pass the data collection to be stored.

This field also affects the online visualization. You can only display characteristics that are processed by the logic of the PDV data collection and are not only stored.

Formula parameters only

Use this option to define that the created process parameter is only used as parameter to calculate a further process parameter. If you set this option for a process parameter, this process parameter will never assess measured values with respect to violated limit values and these process parameters will never be stored in the database.

Compute limit values

If you enable this option, the fields for the calculation of target values or limits are activated. Formula calculation, see section "Formula calculation"

Upper TL formula

You can use the formula for the calculation of the UTL to calculate a new UTL in combination with other parameters. If you recalculate the UTL, this will also lead to changed target values. See section 3.5 for further information on formulas.

Upper PAL formula

You can use the formula for the calculation of the UPAL to calculate a new UPAL in combination with other parameters. If you recalculate the UPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Target value formula

You can use the formula for the calculation of the target value to calculate a new target value in combination with other parameters. If you recalculate the target value, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower PAL formula

You can use the formula for the calculation of the LPAL to calculate a new LPAL in combination with other parameters. If you recalculate the LPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower TL formula

You can use the formula for the calculation of the LTL to calculate a new LTL in combination with other parameters. If you recalculate the LTL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Formula calculation

Using the formula calculation function, you can calculate the measured values collected via machine connection before you evaluate them. For example, you can add an offset to the calculation of a characteristic, you can directly convert units in the collection process or you can perform calculations in combination with other characteristics.



To store and display a formula, the user must have the function authorization `iriscp.formula`. If the user is not authorized, the field is not displayed.

You specify the calculation of measured values in the application *Characteristics* in tab *General*. You can also calculate the target values of an automatically collected characteristic and the upper/lower tolerance or action limits.

Important: HYDRA versions below MES Weaver 2.0 with server systems AIX, HP-UNIX, SCO-UNIX or DEC ALPHA do not provide this kind of formula calculation.

The first part of the formula specifies the level where the formula calculation is made. The following types are available:

- V - Calculation on the level of single values without self-reference.
For each measured value of the characteristics involved, exactly one single value is generated for the calculated characteristic.
- O - Calculation on the level of samples with self-reference.
Use this type of formula to refer to the value itself and to integrate limit or target value.

After the above identifier, the actual formula is specified. The following operators, functions and constants are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
$x + y$	Addition
$x - y$	Subtraction
x / y	Division
$x * y$	Multiplication
$x ** y$	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

The following syntax `[A:B:C]` applies for the variables that identify the single or default values of the process parameters involved. The available values are listed below.

You can specify the following values for section A:

- X – single value/measured value
- UTL – Upper Tolerance Limit
- UPL – Upper Process Action Limit
- TV – Target Value
- LPL – Lower Process Action Limit
- LTL – Lower Tolerance Limit

Note: The value X defined for section A (single value/measured value) is only used for the calculation of measured values; this means, the value is not used in the target value formulas or formulas of the upper/lower tolerance or action limits.

Note: Aggregate functions like MAX, MIN or AVG, which are used in HYDRA CAQ, are not supported with automatically collected characteristics.

Section B describes how the relevant characteristic is identified. The following possibilities are available:

- SELF – self-reference
(This requires formula type O)
- PPARAM – Reference to further process parameters of the same machine
With this formula, only process parameters of the same machine can be calculated.

Section C identifies the characteristic using the field content specified in section B. This means: If a self-reference is specified in section B, a section C may not exist. If instead a reference to a further process parameter is stored for PPARAM, the process parameter must be specified in section C.

Example 1:

The process parameter always collects ten times the rounded value from the machine connection.

→ Formula: O: `round([X:SELF]) * 10`

Example 2:

The characteristic "area" results from the product of the process parameters LENGTH (LAENGE) and WIDTH (BREITE). For each single value of the two source characteristics, a single value is calculated for the characteristic "area".

→ Formula: V: [X:PPARAM:LAENGE] * [X:PPARAM:BREITE]

Example 3:

The following formula is not stored for the measured value, but for the target value of the process parameter "speed". The target value is calculated using the own target value plus the target value of process parameter DURCHMESSER (diameter). The target value for "speed" is therefore recalculated, if the target value for "diameter" changes.

→ Formula: O: [TV:SELF] + [TV:PPARAM:DURCHMESSER]

Note:

The formula calculation is performed directly after reception of the measured values from the PDV data collection. Changes to any of the process parameters included in the formula are then used as events that trigger recalculation.

4 Summary

4.1 General notes on the document

This document describes the "Groups", e.g. article groups, application of the Manufacturing Operation Center (MOC). For general information on how to use MOC, please refer to the "[moc_cc.pdf](#)" document.

5 Groups

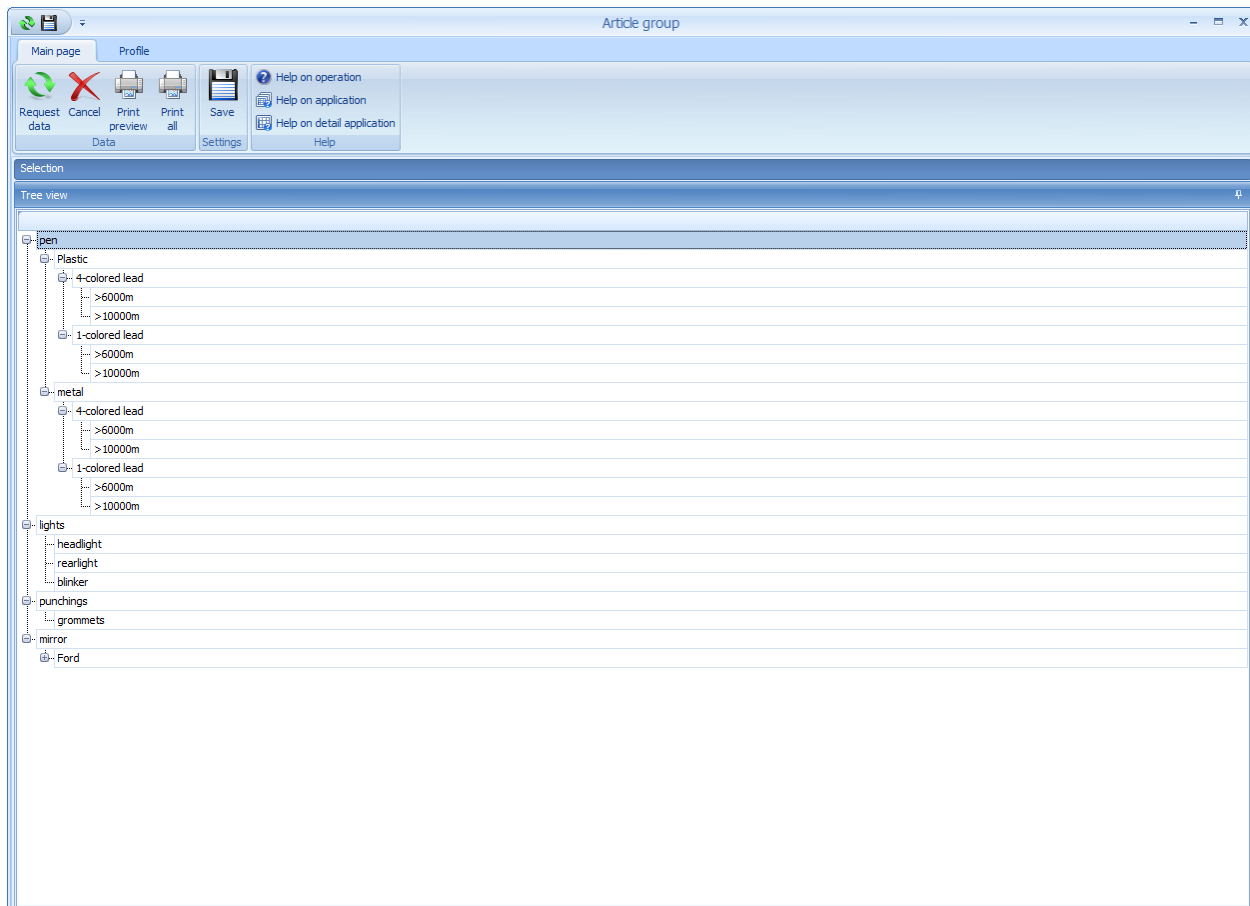
The group catalogs have been designed to create and edit groups for the different applications. The created groups may be assigned to master data of the corresponding application. Consequently, article groups may be created, for example, and assigned to the articles. In this case, it is also possible to create inspection plans on the basis of article groups.

Basically, the creation of groups is also reasonable for failure mode analyses.

5.1 Starting the function

Menu	Master data → Quality management → Article groups Master data → Process data processing → Article groups Master data → Quality management → Measure groups Master data → Quality management → Failure type groups Master data → Quality management → Failure location groups Master data → Quality management → Failure cause groups Master data → Quality management → Causer groups Master data → Quality management → Cost type groups
Transaction code	atcgr → Article groups measgr → Measure groups ftypgr → Failure type groups flocgr → Failure location groups fcaus → Failure cause groups origr → Causer groups costgr → Cost type groups
Function authorization	atcgr -> Article groups measgr.* → Measure groups ftypgr.* → Failure type groups flocgr.* → Failure location groups fcaugr.* → Failure cause groups origr.* → Causer groups costgr.* → Cost type groups

5.2 Default Application Layout



This figure of the article group catalog is exemplary for all groups.

5.3 Toolbar

The toolbar contains the function calls that are available for this application and possibly links to other applications. The functions placed on the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions such as “help”, “request data”, “save application settings” and “print preview”, the other tabs include specific functions that are tailored to the corresponding detail application. The individual functions of the application are listed in the paragraphs that follow.

"Data" category



Request data

The information to be displayed within the application is requested according to the entered selection criteria. This process might take some time depending on the data set from which the system filters data and on the selection result to be transferred and displayed.



Cancel

This function cancels the query sent by clicking the “request data” button.



Print

preview

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.



Save

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking “Yes”.

"Functions" category

There are no special functions for this detail application. Groups are created, changed and deleted using the context menu of the right mouse button within the tree structure view.

"Help" category



Help on operation

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled “moc-cc.pdf”. It describes how to use MOC in general and applies for all applications.



Help on application

This function opens the manual that describes the application from which the help function was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

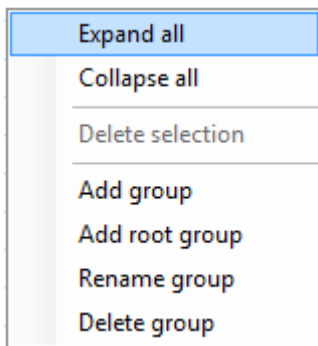
This function opens the application manual at the section where the respective detailed application is described.

5.4 Selection parameters

There are no selection parameters. A specific group can be found by using the “fast selection” function. To use the “fast selection” function just open the group tree structure on the 1st level, select the first entry and enter the first letter of the group in question. Consequently, the first group starting with this letter that is found is selected. The “fast selection” function also integrates subordinate groups that are not opened. If the requested term is included in a group that is not opened, it will be opened automatically.

5.5 "Groups" Detail Application

A group may be created, changed or deleted by opening the context menu of the right mouse button within the display area of the tree structure.



It is altogether possible to define groups up to the fifth hierarchy level. The “add root group” function has to be selected in the context menu of the group tree to create a new main group (1st level). The menu entry “add group” generates a sub-group (level 2 to 5). A designation, which is directly entered in the list view, has to be assigned for this new group. To be able to save the new group, click above or below this new entry within the group tree. Then a confirmation prompt appears asking whether or not the new group is to be saved. Provided that this question is affirmed (“yes”), the entry is saved. The same applies for renaming of groups. Regardless of which hierarchy level is concerned, the entry has to be selected to be able to edit the group designation. Changes can directly be entered in the corresponding line of the tree view. Click above or below the entry to be changed to be able to save the modification. No confirmation prompt appears when it comes to renaming.

An entry is also deleted by selecting a group entry and executing the “delete group” function. Only the group that is at the bottom of the group tree can be deleted.

The “expand all” context menu entry opens all groups up to the lowest hierarchy level. The “collapse all” context menu option closes all entries up to the first level.

The "delete selection" function cannot be used in the maintenance of groups dialog. This function is enabled, for example, in the maintenance of articles application if an article group is selected and this selection is to be removed/deleted.

6 Article

This document describes the "article" application of the Manufacturing Operation Center (MOC). General information on how to use MOC can be found in the document entitled "[moc_cc.pdf](#)".

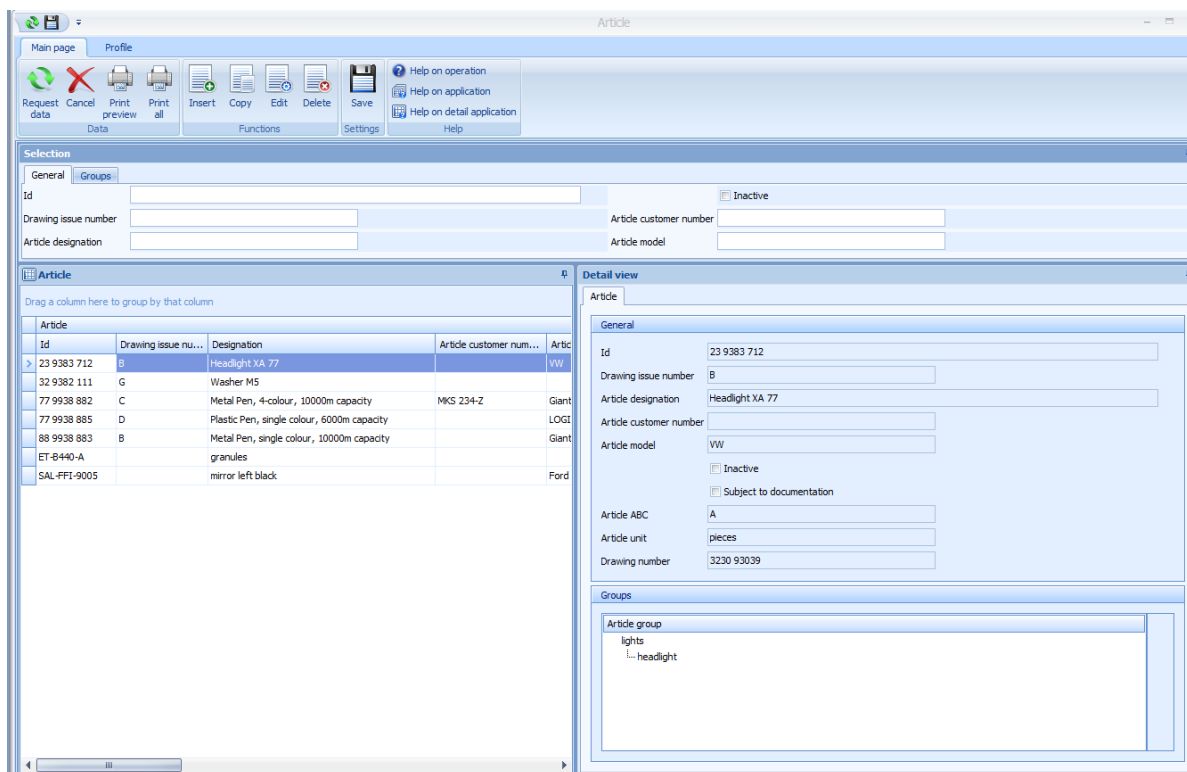
The article catalog has been designed to edit/keep articles. Article data is a global catalog that is used in many CAQ modules and in PDV (Process Data Collection). Provided that there is an interface to a higher-level system (e.g. ERP system), articles may be created automatically via this interface. As soon as a new article is created or changed, e.g. in the ERP system, the article data record is automatically created or changed in the HYDRA-CAQ article catalog based on the defined information.

6.1 Function call

Menu	Master data → Quality management → Article Master data → Process data processing → Article
Transaction code	atc
Function authorization	atc

Available user fields		
Location	Object type/user field key	Source (type)
Table and detail view	ATK/SYSTEM	MF-D
How can I configure user fields? Which user field types are available?		

6.2 Default Application Layout



6.3 Toolbar

The toolbar provides the different functions available for this application and possibly links to other applications. The functions included in the “general” tab of the toolbar are available in all detail applications. In addition to the standard functions, such as help, request data, save application settings, and print preview, the other tabs also include specific functions that are specially tailored to the respective detail application. The following sections describe the individual application functions.



Request data

The information to be displayed within the application is requested on the basis of the entered selection criteria. This process might take some time depending on the dataset from which the system filters data and on the selection result to be transferred and displayed.



Cancel

The query sent by clicking the “request data” button can be canceled using this function.

**Print preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

Category Functions

**Add**

Adds a new article.

**Copy**

Copies the selected article.

**Edit**

Edits an already existing article

**Delete**

Deletes the selected or several selected articles.

Category Help

**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "[moc_cc.pdf](#)". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual for the respective application from which the help file was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

This function opens the application manual at the section where the relevant detailed application is described.

6.4 Selection parameters

The application provides the following selection criteria:

Tab "General"

- Article no.:
Article number
- Drawing issue number:
Drawing issue number of the article, often also referred to as index
- Designation:
Article name
- Inactive:
Inactive, active articles. The checkbox is not enabled by default.
- Customer article no.:
Customer article number
- Article model:
Article model

Tab "Groups"

- Group:
The article group tree can be opened using the  button if an article group is to be filtered. There is a function to accept and cancel the activity.

6.5 Detail application "Article"

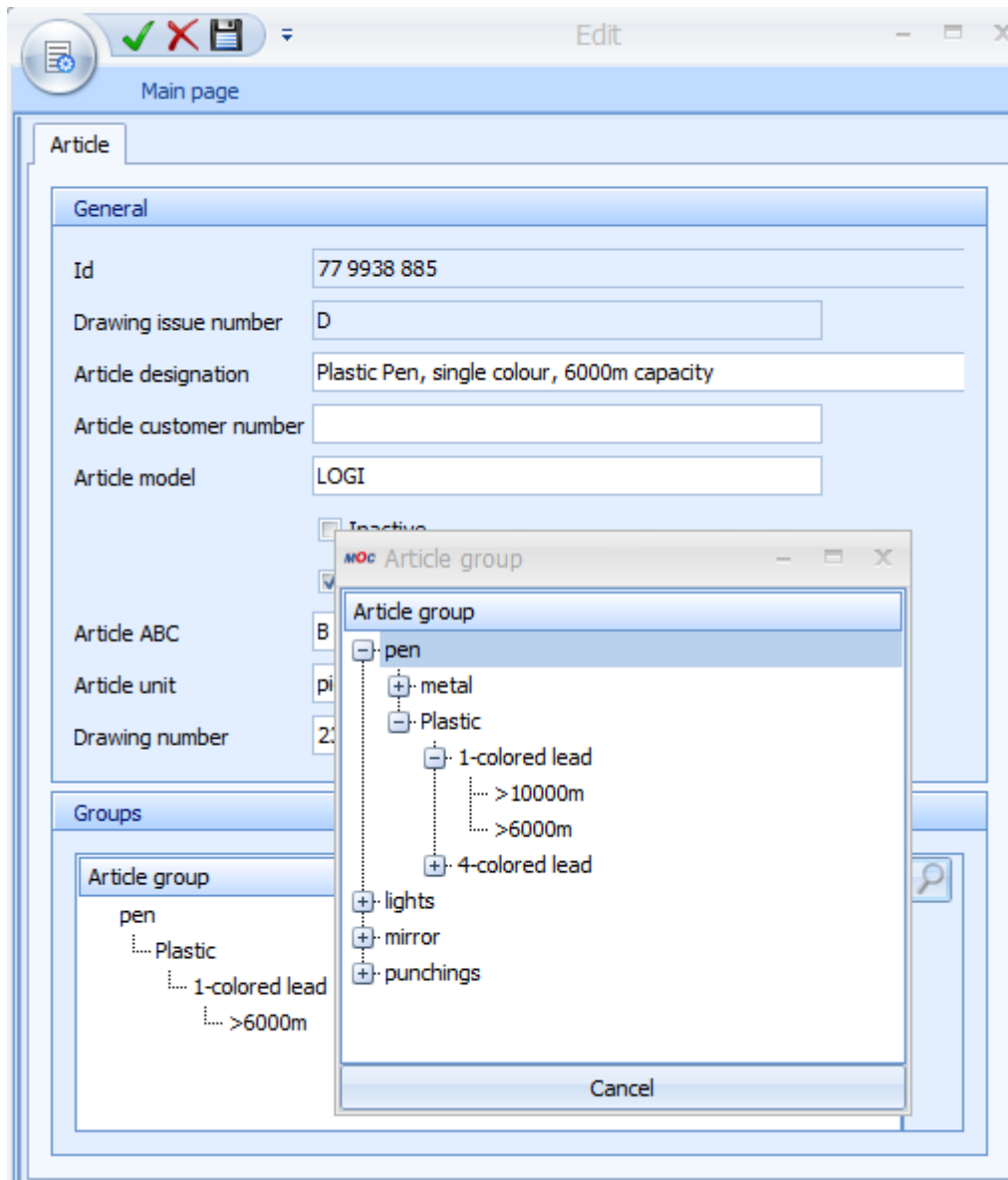
The article number as well as the drawing issue number uniquely identify articles in all areas of HYDRA-CAQ referring to the article catalog. The drawing issue number, also referred to as article index, may be very important, in particular, for inspection planning and when inspection orders are generated. Thus, it is, for example, possible to create an inspection plan for the article 12938 with the drawing issue numbers A and B. Different inspection specifications apply for each drawing issue no. Unless the drawing issue number is indicated and thus may be part of the inspection plan, the system that generates the inspection requirements, must deliver this drawing issue number.

The fields “article no.” and “drawing issue number” fields are key fields, i.e. when a new article is saved, it is first checked whether an article with this key information already exists.

By distinguishing between active and inactive articles, it may be defined whether or not the articles are available in certain selection lists. Thus, no inspection plan can be created for an inactive article. However, inactive articles may be evaluated at any time. Moreover, inactive articles can also be reactivated at any time.

Furthermore, an article can be defined as being subject to documentation. In addition the dialog provides the fields customer article number, article model, article ABC, drawing number as well as the possibility to assign units. To assign units (dimensions), the catalog of units is used.

If you want to make evaluations on article groups or if you use family inspection plans, it is mandatory to assign the respective group. To assign groups, open the group tree using the lens icon. Using the hierarchic tree entries the required group can be selected in the group tree and accepted by double clicking.



The assigned group including the hierarchical group structure then appears in the “groups” field of the editing dialog of articles.

When articles are displayed in a list, the group hierarchy is represented by the columns “group 1 to group 5”.

Groups are maintained in the “article groups” application and is described in the document entitled [“MOC_Groups.pdf”](#).

7 PDV Specification List

Overview

Menu	Master data → Process data processing → PDV specification list
Transaction code	scip
Function authorization	scip

Purpose

The specification list has been designed to create specifications, e.g. within the framework of family inspection planning.

Set the option "from list" in the relevant inspection plan characteristics to use the specifications of the specification list when generating inspection steps with inspection step characteristics. In this case, only active entries of the specification list are used. The application uses the following key fields to search for a specification list entry, when generating an inspection step.

- Area
- Machine no.
- Article number
- Resource number
- Characteristic no.
- Operation number and operation designation (name)

You can configure the order for searching the specification list while customizing the system.

The specification list does not replace inspection planning. It rather is a supplement to family inspection plans.

Integration

This function is a fundamental component of family/group inspection planning. The specification list defines the inspection specifications that vary with each item/article in the article group of a group inspection plan.

The function is largely identical to the CAQ specification list. With the exception that in the PDV module you can only select "article-related PDV inspection planning" as available area.

Requirements

An inspection plan for the article group including corresponding characteristics referring to the specification list has to exist in order to use the specification list (configure the option "from list" in the inspection plan characteristics).



Please note that sample-related reports/evaluations are not available for process characteristics whose limit values are controlled via specification lists.

Selection criteria

The application provides the following selection criteria:

Area

Shows the available CAQ areas. The PDV module only supports the "article-related PDV inspection planning".

Specification no.

Unique specification number.

Version no.

Unique version within the specification. Only active, provided that you have activated the version control function in the system settings.

Active

You can filter by active or inactive entries.

Article number

Article number of the specification list entry. You can select the article number from the catalog for article master data.

Article name/designation

Name of the article.

Operation

Operation number.

Workplace

Workstation, e.g. machine.

Characteristic no.

Characteristic number.

Characteristic designation/name

Characteristic designation/name

If you select multiple selection criteria, the specification list shows the matching results.

Field descriptions***Characteristic key tab******"Characteristic" group*****Area**

Specifies the area for which the specification list entry is to apply.

Specification no.

Number of the specification.

Version no.

Version number of the specification.

Active

Checkbox. Shows whether the entry is active or inactive.

Workplace

Indicates the workplace. If you enter a workstation (e.g. machine) you can check articles/items with respect to this workplace. This might be useful if you use machines of different types.

Machine name

Designation/name of the workplace/machine.

Resource

You can enter a resource (e.g. tool).

Designation (name)

Name of the resource.

Characteristic no.

Unique number of the characteristic.

Characteristic designation/name

Name of the selected characteristic number.

Operation

Number of the operation.

Operation designation/name

Designation/name of the operation.

Article number

Article number for this specification list entry.

Article name/designation

Designation/name of the selected article number.

Drawing issue number

Drawing issue number of the selected article number.

***Specifications* tabs and other tabs**

The remaining index tabs are described in the characteristics definition master data application.

Toolbar

The below-mentioned additional functions are available besides the standard functions.

**Activate**

Function authorization: sclp.active

Click this button to activate a specification list entry. This function automatically disables a previously released version.

**Deactivate**

Function authorization: sclp.release

Click this button to deactivate a specification list entry. The specification list entry is no longer used.